ZIYANG CHEN

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EDUCATION

University of Michigan, Ann Arbor Jan. 2022 -
Ph.D. in Electrical & Computer Engineering | Advisor: *Andrew Owens*

University of Michigan, Ann Arbor Sept. 2019 - May 2021
Master of Science in Engineering, Computer Vision
Overall GPA: **4.0/4.0** | Advisor: *Andrew Owens*

Shanghai Jiao Tong University, UM-SJTU JI Sept. 2015 - Aug. 2019
Bachelor of Science in Engineering
Major: Electrical & Computer Engineering | Minor: Entrepreneurship
GPA: **3.60/4.0** with honors & *Graduated magna cum laude*

RESEARCH EXPERIENCE

Stereo Audio Contrastive Random Walk June 2021 - Present
Research Associate, University of Michigan *Advisor: Andrew Owens, David Fouhey*

- Proposed a self-supervised method to learn interaural time difference in stereo audio with contrastive random walk on the internet videos
- Applied learned ITD signal to localize the sound sources in the images
- Evaluated on a synthetic dataset with pseudo ITD ground truth and achieved significantly better results

Audio Analogies June 2021 - Present
Research Associate, University of Michigan *Advisor: Andrew Owens, Bryan Russell, Justin Salamon*

- Proposed an audio-visual synchronization method conditioning on given videos
- Applied our proposed method to audio-visual forensics downstream tasks to detect fake videos

Structure from Silence Jan. 2020 - May 2020, Apr. 2021 - Sep. 2021
Research Associate, University of Michigan *Advisor: Andrew Owens*

- Collected datasets with ambient sound and paired RGB-D videos of the indoor environment
- Demonstrated that the ambient sound could infer geometric properties with experiments and proposed a new audio-visual learning method to learn exploit structural representation
- Apply learned models to robots and navigate it via ambient sound only in the real world *(CoRL 2021)*

Semantic Forensics, DAPRA SemaFor Jan. 2021 - Present
Research Associate, Mentor, University of Michigan *Advisor: Andrew Owens*

- Implemented real-or-fake image detection and rebuilt datasets with samples from different generators
- Designed experiments about fake image classification to solve the generator attribution problem
- Proposed a zero-shot fake face detection method by learning the consistence in real images
- Attended the Hackathon and solved fake image detection challenges

Cycle-Consistent Audio-Visual Grouping July 2020 - Present
Research Associate, University of Michigan *Advisor: Andrew Owens*

- Implemented state-of-the-art sound localization methods and grouping algorithms as baselines
- Proposed a simple self-supervised audio-visual method using contrastive random walk with cycle-consistency constraints and applied learned representations to sound source localization and grouping tasks
- Evaluated our method on human speech and musical instruments datasets and achieved significantly better qualitative and quantitative results *(In Submission)*

Application of Ad-Words Problem under Carpool Situation Sept. 2019 - Jan. 2020
Research Associate, University of Michigan *Advisor: Cong Shi*

- Applied primal-dual online match algorithms in practical application
- Utilized mathematics derivation to prove the advantage of the proposed algorithm

WORK EXPERIENCE

Research Assistant, *University of Michigan, Ann Arbor* Jan. 2020 - Present

- Topics: *audio-visual representation learning*, *self-supervised learning*

Teaching Assistant, *Shanghai Jiao Tong University* Sept. 2018 - Aug. 2019

- Courses: *Digital Integrated Circuits*, *Electronic Circuits*
- Conducted experiments, held office hours and taught recitation classes every week

Research Engineer Intern, *FreeSense Image Inc., Shanghai* Mar. 2018 - May 2018

- Implemented an auxiliary system for a project of Detection of Car LCD Screen and applied it into industrial production lines
- Adopted state-of-the-art recognition DNNs to improve the accuracy on defect detection tasks

SELECTED PROJECTS

Prediction Model based on Bayesian Hierarchical Clustering *SJTU*

- Processed data, re-generated and adopted efficient algorithms in Julia
- Applied Bayesian Hierarchical Clustering algorithm to make roughly correct prediction on center point of clusters with limited data provided

Design of Evaluative System for Online Car Hiring Service *SJTU, DiDi Co.*

- Applied China-Word-Vector & Recurrent Attention Network to train the deep learning model which can capture opinions and analyze sentiments
- Built a website in full stack with Python Flask and HTML+CSS/JS to provide online services

Driver's Alarming System in Vehicle *SJTU*

- Performed face keypoint detection with OpenCV and predicted whether the driver is sleepy or not
- Embedded our alarm system in Raspberry Pi with a camera installed

PUBLICATIONS

- *Ziyang Chen*, Xixi Hu* and Andrew Owens. Structure from Silence: Learning Scene Structure from Ambient Sound. (CoRL 2021, Oral)*
- *Xixi Hu*, Ziyang Chen* and Andrew Owens. Cycle-Consistent Audio-Visual Grouping. (In Submission)*

AWARDS & SCHOLARSHIPS

- **Lums Fellowship**, *University of Michigan, Ann Arbor* 2019-2021
- **Outstanding Graduate - Class of 2019**, *Shanghai Jiao Tong University* 2019
- **Academic Excellence Scholarship**, *Shanghai Jiao Tong University* 2016-2018
- **Dean's List**, *Shanghai Jiao Tong University, SJTU-UM JI* 2015-2018

OTHERS

Skills

- Programming: Python, C/C++, Matlab, Julia, JavaScript, HTML/CSS
- Tool: OpenCV, PyTorch, TensorFlow, LaTeX, Docker